

Knowledge Map

What I Know, What's Known, and What's Open

Patrick S. Mahon

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Abstract

A consolidated reference covering everything explored in this research programme: what I understand, what is known (and by whom), what I proved that turned out to be classical, what remains genuinely open, and where the understanding might lead. Not a paper. A map for finding the next problem.

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1 The central question

When does observationally coherent data determine a genuine probability measure, and what additional commitments does this require?

This decomposes into three layers:

1. **Extension:** compatible finitely additive charges on a directed system of Boolean algebras $\rightarrow$ σ -additive probability measure.
2. **Realization:** measure on the dual space (Stone space) $\rightarrow$ measure on an underlying state space Ω .
3. **Value-definiteness:** probabilistic description $\rightarrow$ simultaneous definite values for all observables.

Each layer requires a commitment the previous does not. The algebraic structure of the observation algebra constrains which commitments are available; distributivity is sufficient to make all three available.

2 Layer 1: From charges to measures

2.1 The classical results (all known)

- **Continuity from above:** A finitely additive charge on an algebra is σ -additive iff $E_n \downarrow \emptyset \Rightarrow \ell(E_n) \rightarrow 0$ [17, §13, Theorem D].
- **Carathéodory extension:** A σ -additive premeasure on a semiring extends uniquely to a measure on the generated σ -algebra [17, §13], [5, Vol. 1, Ch. 1].
- **Kolmogorov extension:** Compatible finite-dimensional distributions on a product space extend to a unique probability measure, under standard Borel / tightness conditions [4, Ch. 36].
- **Choksi inverse limits:** Compatible measures on an inverse system of compact Hausdorff spaces extend to the inverse limit [9, Theorem 1].
- **Yosida–Hewitt decomposition:** Every bounded finitely additive charge decomposes uniquely as $\ell = \ell_c + \ell_p$ where ℓ_c is σ -additive and ℓ_p is purely finitely additive [41].

2.2 What I proved (turned out classical)

Collective exhaustion (CE) in a directed system: defined as $\exists j \geq i$ such that $\ell_j(\pi_{ij}^{-1}(E_n)) \rightarrow 0$. Under compatibility, the existential witness is vacuous: $\ell_j(\pi_{ij}^{-1}(E_n)) = \ell_i(E_n)$ for all $j \geq i$. So CE = σ -additivity of each ℓ_i . This is not a theorem; it is the observation that the definition contains a redundant quantifier.

The directed-system extension theorem (Paper I, Theorem 2.3) is Kolmogorov extension in directed-system language with “sequential common refinements” replacing the usual product structure.

The Lean formalization (0 sorry on the Carathéodory route, 1 sorry for Yosida–Hewitt on the Stone route) is genuine verified infrastructure, even though the theorems are classical.

2.3 What I understand from this

1. The passage from finite additivity to σ -additivity is the critical non-finitary step in probability. Compatibility, normalization, and finite additivity are all first-order / finitary conditions.
2. No first-order condition can force σ -additivity. Immediate from Łoś's theorem: the ultraproduct of Dirac masses on $\mathbb{N}$ is a purely finitely additive charge [20, 12].
3. The finite-cofinite charge on $\mathbb{N}$ is the canonical obstruction: normalized, finitely additive, not σ -additive, and arises as the ultraproduct witness.
4. The Yosida–Hewitt decomposition gives the exact characterization: $\text{CE} \Leftrightarrow \ell_p = 0$.

3 Layer 2: From dual space to realization

3.1 The classical results (all known)

- **Stone representation:** Every Boolean algebra is isomorphic to the clopen algebra of a compact, Hausdorff, totally disconnected space (its Stone space) [22].
- **Charge $\leftrightarrow$ measure on Stone space:** Every finitely additive charge on a Boolean algebra B corresponds to a regular Borel (Baire) measure on its Stone space $\text{St}(B)$ (the space of ultrafilters on B) [13, §326], [17, §53].
- **σ -additivity $\leftrightarrow$ support:** A charge on B is σ -additive iff the corresponding Stone-space measure vanishes on meagre Baire sets, equivalently assigns full mass to countably complete ultrafilters [32, Theorem 10.5.3].
- **Principal ultrafilters:** Every principal ultrafilter is countably complete. The converse holds for $\mathcal{P}(\kappa)$ when κ is below the first measurable cardinal (Ulam's theorem). For non- σ -complete algebras (typical cylinder algebras), countably-complete non-principal ultrafilters can exist without large cardinals.

3.2 What I proved (turned out classical)

Stone measure exists unconditionally: Any compatible charges produce a Baire probability measure $\hat{\mu}$ on $\text{St}(\mathcal{C})$. This is Stone representation + Choksi. Lean-verified (0 sorry).

Descent requires $\hat{\mu}(\text{pure}(\Omega)) = 1$ (where $\text{pure}(\omega) = \{E \in \mathcal{C} : \omega \in E\}$ is the principal ultrafilter at ω): concentration on principal ultrafilters. Under discriminability and the assumption that all countably-complete ultrafilters in the relevant subalgebra are principal (which holds for σ -complete algebras below the first measurable cardinal), this is equivalent to σ -additivity. The support characterization is Rao–Rao 10.5.3; the principal-vs-countably-complete gap is the subtlety.

Realization is unconstrained: Any surjective-projecting $S \subseteq \text{St}(\mathcal{C})$ can serve as Ω . This is the abstractness of Boolean algebras recognized since Stone (1936).

3.3 What I understand from this

1. The Stone construction gives the *unconditional* object: $\hat{\mu}$ always exists. It is the “empirically adequate” object in van Fraassen’s sense.
2. Descent to Ω is a *choice*: which ultrafilters count as “realized” is additional structure over the algebra.

3. The three-layer nesting:

$$\text{pure}(\Omega) \subseteq \{\text{countably complete ultrafilters}\} \subseteq \text{St}(\mathcal{C}).$$

σ -additivity gives mass on the middle set (intrinsic). Descent needs mass on the inner set (depends on Ω).

4. The van Fraassen identification (Stone measure = empirical adequacy (EA), descent = realist commitment) has no precedent in the literature. It is a philosophical contribution, not a mathematical one.

4 Layer 3: From probability to value-definiteness

4.1 The classical results (all known)

- **Boolean algebras admit 2-valued homomorphisms:** Ultrafilters on Boolean algebras are exactly the 2-valued homomorphisms. Existence by Zorn. *Stone (1936); every Boolean algebra textbook.*
- **Kochen–Specker:** For the lattice $L(H)$ of closed linear subspaces of a Hilbert space H with $\dim H \geq 3$, no dispersion-free state (2-valued homomorphism) exists [25].
- **Gleason:** For $L(H)$, $\dim H \geq 3$, every σ -additive state is $s(P) = \text{tr}(\rho P)$ [15].
- **Bub–Clifton:** Given a state ρ and a preferred observable R , there is a unique maximal Boolean subalgebra $D(\rho, R) \subseteq L(H)$ on which the state is a mixture of dispersion-free states. [7, 18].
- **Orthomodular lattice (OML) duality:** The category of OMLs is dually equivalent to a category of orthomodular spaces. Principal filters $P(A)$ are part of the dual structure (unlike Stone duality where they are invisible). [29]. Zero citations in quantum foundations as of 2026.
- **Colimit generation:** Gluing Boolean algebras via categorical pushout produces orthomodular lattices, generically non-distributive. [16].

4.2 What I proved (classical + one novel framing)

Commensurability theorem: Distributivity is sufficient for value-definite realism (VDR); for $L(H)$, $\dim H \geq 3$, non-distributivity blocks it. Lean-verified (0 sorry, 1 axiom for Kochen–Specker (KS)). *The mathematics is the assembly of Stone + KS + Gleason + Bub–Clifton. Not a new theorem.*

The EA / probabilistic realism (PR) / VDR vocabulary connecting van Fraassen’s empirical adequacy to quantum logic via McDonald–Bimbó duality is novel framing. No precedent found.

The coherence filtration $\mathcal{S}(A) \supseteq \mathcal{S}_\sigma(A) \supseteq \mathcal{S}_{\text{df}}(A)$ with three named transitions (pasting, regularity, sharpness) organizes known results. The chain itself is textbook (Pták–Pulmannová, Kalmbach); the transition names and the observation that distributivity collapses the filtration are new packaging.

4.3 What I understand from this

1. **Distributivity controls value-definiteness.** Boolean algebras have ultrafilters (VDR available). Non-Boolean OMLs ($\dim \geq 3$) do not (KS). The dividing line is distributivity — sufficient but not necessary in full generality ($\dim 2$, horizontal sums).

2. **The algebra can constrain realization in the OML case.** Unlike Boolean Stone duality, the McDonald–Bimbó dual carries $P(A)$ as structure. “Realization is additional structure” in the Boolean case; it is *constrained* structure in the OML case.
3. **The realism debate has different mathematical answers for different algebras.** Boolean: EA, PR, VDR all available; philosophical choice. OML ($\dim \geq 3$): VDR blocked; PR available (Gleason); EA always available.
4. **The topos programme is more powerful.** Isham–Butterfield / Döring–Isham / Heunen–Landsman–Spitters encode the same structure via spectral presheaves and Bohri-fication. Our approach is coarser (single dual space, not presheaf over contexts) but connects more directly to van Fraassen.

5 Delay reconstruction

5.1 The classical results

- **Takens:** Generic smooth observable gives an injective delay map for $L \geq 2 \dim X$ [39].
- **Sauer–Yorke–Casdagli:** Prevalence-based embedding theorem [35].
- **Generating partitions:** $h(T, P) = h(T)$ iff P generates [37, 26].
- **Grassberger–Procaccia:** Correlation integral estimates D_2 and K_2 (Rényi-2 dimension and entropy) from delay data.
- **False nearest neighbours (FNN):** Embedding dimension selection [24]. Sensitive to noise [33].
- **Ragwitz–Kantz:** Local prediction error minimization for (d, L) selection [31].
- **Botvinick–Greenhouse et al.:** Measure-theoretic Takens via optimal transport [6].

5.2 What I proved (mostly classical)

Reconstruction theorem: Observable σ -algebra = full σ -algebra mod μ iff delay algebra dense in L^2 iff delay map is measure-theoretic embedding. This is Krieger’s generating partition theorem [26] in delay-map language.

Predictive factorization, Chapman–Kolmogorov, Koopman–Perron: All classical (Rokhlin, Kolmogorov, Koopman 1931).

Finite-sample convergence rates: Minimax rates $n^{-s/(2s+d)}$ [38]; elbow stopping via Lepski [27]. Not cited in the original work — a significant attribution gap.

5.3 What is genuinely novel (but motivation unclear)

Geometric $\neq$ algebraic reconstruction (2026-05-14): The bridge note claimed $\delta(L) \rightarrow 0 \Leftrightarrow (\mu \otimes \mu)(R_L) \rightarrow 0$ under a fibre mixing condition. This is **false**: the core identity (Lemma 2.1 of the bridge note) has a disintegration error. The conditional variance $\mathbb{E}[\text{Var}(\mathbf{1}_S \mid \mathcal{O}_L)] = \sum_z p_z \mu_z(S) \mu_z(S^c)$ is weighted by p_z , while the collision integral $(1/2) \int_{R_L} |\mathbf{1}_S - \mathbf{1}_{S'}|^2 d(\mu \otimes \mu) = \sum_z p_z^2 \mu_z(S) \mu_z(S^c)$ is weighted by p_z^2 . The bridge note equated them; they are genuinely different.

Skew-product counterexample: $X = A^{\mathbb{Z}} \times B^{\mathbb{Z}}$, $T = \sigma \times \sigma$, $h(a, b) = a_0$, independent iid uniform. Then collision = $2^{-(L+1)} \rightarrow 0$ (fibres shrink in the a -coordinate) but $\delta_L = 1/4$ for all L (the event $\{b_0 = 0\}$ is maximally unresolved). This falsifies even the “easy direction.”

What survives: The definitions (collision probability, Rokhlin distance, fibre mixing) are well-posed. The trivial direction $\delta \rightarrow 0 \Rightarrow \text{collision} \rightarrow 0$ holds. The converse and the entropy characterization $\delta \rightarrow 0 \Leftrightarrow H_2 \rightarrow \infty$ are both dead.

Replacement direction: A short note proving that geometric and algebraic reconstruction are inequivalent, with the skew-product as the central theorem. The open question becomes: what conditions on the factor map make $\text{collision} \rightarrow 0$ imply $\delta \rightarrow 0$?

5.4 What I understand from this

1. The reconstruction problem is *algebraically underdetermined*: the observation algebra does not constrain the realization. Embedding selection (d, L) is necessarily empirical.
2. Every practical reconstruction criterion (FNN, mutual information, Lyapunov convergence, prediction error) implicitly assumes a modelling commitment. Deterministic criteria fail under noise because noise violates the commitment.
3. The Rokhlin distance $\delta(\mathcal{O})$ is the measure-theoretic notion of “how close is the observable σ -algebra to generating everything.” It is the Rokhlin metric, named and studied since 1967.
4. Collision probability / Rényi-2 entropy measures the “collision structure” of the fibre decomposition. Grassberger–Procaccia already extracts this from data.
5. The bridge theorem linking collision convergence to σ -algebra generation is false: the disintegration error (p_z vs p_z^2 weighting) means the proposed identity is simply the wrong identity, and the inequivalence of geometric and algebraic reconstruction is known/obvious.

6 Dead ends and lessons

1. **De Finetti via Stone:** Exchangeability forces σ -additivity (Hausdorff moments). The “exotic part” of the Stone measure is empty. Dead end.
2. **Stationary Stone:** The Yosida–Hewitt decomposition commutes with shift-invariance. An observation, not a theorem. The real open question (ergodic theory of the purely finitely additive part) is out of reach.
3. **Dynamics from observation:** “Can the Stone space give rise to dynamics without positing T ?” No: the algebra doesn’t constrain Ω , so it certainly doesn’t constrain structure on Ω .
4. **Paper III (noise thresholds):** Residual autocorrelation as embedding diagnostic = Billings–Voon [3] / Casdagli [8]. FNN noise sensitivity = Rhodes–Morari [33]. Elbow stopping = Lepski [27]. Rediscovery throughout.
5. **“No intermediate condition”:** The claim that no condition interpolates between first-order and σ -additivity is false. $\limsup \ell(A_n) < c$ trivially interpolates.
6. **Bridge theorem (fibre mixing):** The claimed equivalence $\delta(L) \rightarrow 0 \Leftrightarrow (\mu \otimes \mu)(R_L) \rightarrow 0$ under fibre mixing is false. The core identity has a disintegration error: the conditional variance integral is weighted by p_z , the collision integral by p_z^2 . The skew-product $X = A^{\mathbb{Z}} \times B^{\mathbb{Z}}$, $h(a, b) = a_0$ falsifies even the easy direction. The replacement observation (geometric $\neq$ algebraic reconstruction) is known to ergodic theorists. Both the positive bridge and the negative note fail the audit.

7. **Entropy characterization:** $\delta(L) \rightarrow 0 \Leftrightarrow H_2(\nu_L) \rightarrow \infty$ depended entirely on the bridge theorem. Dead.
8. **Nonlinear semigroups / Hille operator (Phase 1):** C_n -semigroups on Banach spaces, resolvent from finite difference spectra, max-entropy fitting. Abandoned approach — the semigroup formalism was too heavy for the applied question. The Rose kernel was the lighter successor.
9. **Rose–Takens measure conjugacy:** “Takens embedding + $D_{\text{KL}}(h_{\#}\mu\|\nu) = 0$ implies measure conjugacy.” Trivially true by definition: diffeomorphism + pushforward = target measure IS measure conjugacy. Not a theorem.
10. **CGF $\rightarrow$ Fisher $\rightarrow$ Landau chain:** Standard information geometry [2]. The CGF-Fisher connection is the definition of Fisher information.
11. **KL = least action (discrete Onsager–Machlup):** Known: [11], [28], large deviations literature.
12. **Fisher clock / entropy production:** Broken detailed balance $\rightarrow$ entropy production is [36]. The “Fisher clock” framing is poetic but the mathematics is standard nonequilibrium statistical mechanics.
13. **Neutral operator edge law:** The joint edge law $\Gamma = \nu \otimes K$ and its f -divergence structure were absorbed into Paper II (conditional regularity kernel κ_Q). The diagnostic exponents $\eta = 2s$, $\zeta = 2s/(2s + d)$ are the same Lepski/Stone rates already canned in Paper III.
14. **CE via locale theory (Simpson’s sublocale):** Conjectured that $\text{CE} \Leftrightarrow$ Simpson’s smallest measure-1 σ -sublocale is spatial. Dead: δ_U (Dirac at free ultrafilter) gives spatial one-point sublocale but CE fails; compact Hausdorff makes $\Rightarrow$ trivially true for all measures. Non-spatiality is a constructive phenomenon that vanishes classically.
15. **Prediction error $\leftrightarrow$ generation:** Attempted to connect Casdagli’s prediction-error method to σ -algebra generation ($\delta(L)$). Self-deflated: Bayes-optimal prediction error IS conditional variance (definitional); the non-trivial direction (prediction of h implies generation) is just the reconstruction theorem (Paper II).

7 What remains genuinely open

Open Problem 7.1 (Measure-free Boolean algebra). Does there exist a Boolean algebra B satisfying all three conditions:

- (i) B is non- σ -complete (equivalently, $\text{St}(B)$ is not basically disconnected);
- (ii) B is non-atomic (equivalently, $\text{St}(B)$ has no isolated points);
- (iii) B is measure-free: B admits no strictly positive σ -additive probability (equivalently, $\text{St}(B)$ carries no strictly positive Radon probability measure)?

Here σ -additivity on a non- σ -complete B is interpreted for disjoint countable families whose join exists in B .

On the clopen algebra of any Stone space, σ -additivity is vacuous by compactness; the real question concerns Radon measures on the Stone space itself (the Borel σ -algebra generated by clopens). Nearest known constructions (Argyros 1983, under CH; Kunen; Fedorchuk) each miss at least one required condition. Plebanek (2024 survey) confirms the exact parameter regime remains open. Likely ZFC-independent but this is not proved.

Open Problem 7.2 (OML extension). Let A be an orthomodular lattice and $s: A \rightarrow [0, 1]$ a state (i.e. $s(1) = 1$ and $s(a \vee b) = s(a) + s(b)$ whenever $a \perp b$). Let $S_0(A)$ be the McDonald–Bimbó dual space with clopen $\perp$ -stable sets $\text{CO}(S_0(A))^\dagger$, and define $\mu(h(a)) = s(a)$ where $h(a) = \{x \in F(A) : a \in x\}$.

- (a) Under what conditions on A does μ extend to a σ -additive measure on the Baire σ -algebra of $S_0(A)$?
- (b) In particular, does any condition on A strictly weaker than Booleanity suffice?

For $A = L(H)$ with $\dim H \geq 3$, Gleason’s theorem gives the extension: every state is $s(P) = \text{tr}(\rho P)$. For general OMLs, Gleason-type results are unavailable and Carathéodory does not apply (the clopens form an OML, not a Boolean algebra).

The structural obstruction: Pták–Pulmannová (1994) prove that if every unital subadditive measure on an OMP is a state, the lattice must be Boolean. The gap between “too weak” (allows non- σ -additive states) and “too strong” (forces Booleanity) appears structurally robust. Recent incremental work (Bunce–Wright 1992; Chetcuti–Dvurečenskij 2003/2005; Voráček–Pták 2023) confirms no breakthrough. All exotic OML state spaces use Greechie pasting (Wilce 2009, Handbook of Quantum Logic Ch. 24). Needs a new idea, not more reading.

Open Problem 7.3 (Ultralimit representability — reduces to Problem 7.1). Let B be a Boolean algebra, $\mathcal{U}$ a nonprincipal ultrafilter on an index set I , and $(\mu_i)_{i \in I}$ a family of σ -additive probabilities on B . Call a purely finitely additive charge ℓ on B *ultralimit-representable* if $\ell(b) = \lim_{\mathcal{U}} \mu_i(b)$ for all $b \in B$, for some such family and ultrafilter.

For which Boolean algebras B is every purely finitely additive charge on B ultralimit-representable?

Status (2026-05-18): settled in all regimes except Problem 7.1.

- *Atomic B (e.g. $\mathcal{P}(\mathbb{N})$):* Every PFA charge is representable. Ultrafilter charges are realized via Diracs; the rest by Krein–Milman.
- *σ -complete B :* Nikodým convergence forces ultralimits of σ -additive measures to stay σ -additive. No PFA charge arises.
- *Non- σ -complete, non-atomic B :* The canonical decomposition $B \cong (B \upharpoonright a) \times (B \upharpoonright a^c)$ shows that failure of representability is witnessed by a measure-free direct-product factor — reducing to Problem 7.1.

Literature scoped (2026-05-18): Duanmu–Weiss (2018), Cardona et al. (2025), Fremlin §§326–328 checked; none address this question. Not an independent direction.

Open Problem 7.4 (Extreme PFA shift-invariant charges). Let $T: \Omega \rightarrow \Omega$ be a measurable transformation on a standard Borel space. What are the extreme points of the convex set of T -invariant purely finitely additive charges on Ω ?

Connected to the structure of Banach limits and the Yosida–Hewitt decomposition of invariant means. *Out of reach currently.*

8 The understanding, condensed

One sentence per insight:

1. The passage from coherent charges to probability requires σ -additivity, which is not derivable from finitary conditions.

2. The Stone construction gives an unconditional measure on the dual space; descent to a realization requires a commitment the algebra cannot force.
3. Which ultrafilters are “realized” is additional structure in the Boolean case and constrained structure in the OML case.
4. Distributivity is sufficient for value-definiteness; non-distributivity blocks it for $L(H)$, $\dim \geq 3$, but not for all non-distributive OMLs (e.g. $L(\mathbb{C}^2)$, horizontal sums).
5. The realism/empiricism boundary is algebraic: van Fraassen’s empirical adequacy is the unconditional level; value-definite realism requires the algebra to admit dispersion-free states (guaranteed by distributivity, blocked by KS for $L(H)$, $\dim \geq 3$).
6. Delay reconstruction is algebraically underdetermined: embedding selection is necessarily empirical.
7. Geometric reconstruction ($\text{collision} \rightarrow 0$) and algebraic reconstruction ($\delta \rightarrow 0$) are genuinely inequivalent. The bridge theorem claiming their equivalence under fibre mixing was false (disintegration error in the core identity). The skew-product counterexample is definitive.
8. Every practical reconstruction diagnostic implicitly assumes a modelling commitment; noise violates deterministic commitments.
9. Audit before you draft.

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